ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

MATHEMATICS

PRACTICE EXAM 2 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 7 Mathematics practice exam 2.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct answers.

QUESTION 1

1 1.1 D ✓ 1.2 B ✓ 1.3 C ✓ 1.4 A ✓ 1.5 B ✓ 1.6 D ✓ 1.7 A ✓ 1.8 C ✓ 1.9 C ✓ 1.10 A ✓ [10]

QUESTION 2

2.1 $8 \times (100 - 1) = 800 - 8$ ✓ $= 792$ ✓ (2)

2.2 $5 + 3 = 8$ parts; $\text{R}720 \div 8 = \text{R}90$ per part ✓ (3)
Lerato: $5 \times \text{R}90 = \text{R}450$ ✓ Sipho: $3 \times \text{R}90 = \text{R}270$ ✓

2.3 $240 \div 3$ ✓ $= 80$ km/h ✓ (2)

2.4 (a) $\text{R}490 - \text{R}350 = \text{R}140$ ✓ (3)
(b) ${}^{140}/_{350} \times 100$ ✓ $= 40\%$ ✓

2.5 $\text{VAT} = 15\% \times \text{R}800 = \text{R}120$ ✓ $\text{Price} = \text{R}920$ ✓ (2)

2.6 $\text{Interest} = \text{R}2\,000 \times {}^{8}/_{100} \times 3$ ✓ $= \text{R}480$ ✓ $\text{Total} = \text{R}2\,480$ ✓ (3)

[15]

QUESTION 3

3.1 (a) 32 ✓ (b) $5^2 - 9 = 25 - 9$ ✓ $= 16$ ✓ (c) 4 000 ✓ (4)

3.2 $3^4 = 81$ and $4^3 = 64$ ✓, so 3^4 is larger ✓ (2)

3.3 $360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5$ ✓ $= 2^3 \times 3^2 \times 5$ ✓ (2)

[8]

QUESTION 4

4.1.1 ${}^{5}/_{6} - {}^{2}/_{6}$ ✓ $= {}^{3}/_{6} = {}^{1}/_{2}$ ✓ (2)

4.1.2 ${}^{12}/_{5} \times {}^{5}/_{4}$ ✓ $= {}^{60}/_{20}$ ✓ $= 3$ ✓ (3)

4.1.3 ${}^{5}/_{10} + {}^{4}/_{10} + {}^{3}/_{10}$ ✓ $= {}^{12}/_{10} = 1\,{}^{1}/_{5}$ ✓ (2)

4.2 ${}^{125}/_{1000}$ ✓ $= {}^{1}/_{8}$ ✓ (2)

4.3 (a) 3,75 ✓ (b) $9{,}36 \div 4 = 2{,}34$ ✓✓ (3)

4.4 ${}^{3}/_{8} \times 32$ ✓ $= 12$ learners ✓ (2)

[14]

QUESTION 5

5.1 (a) $-9 < -4$ ✓ (b) $-1 > -10$ ✓ (2)

5.2 (a) -14 ✓ (b) -7 ✓ (c) 10 ✓ (3)

5.3 $-18 + 7 = -11$ (11 m below sea level) ✓ (1)

[6]

QUESTION 6

6.1.1 18 ✓ ; 22 ✓ (2)

6.1.2 $T_n = 4n - 2$ ✓✓ (2)

6.1.3 $4(50) - 2 = 198$ ✓ (1)

6.2 48 ✓; multiply each term by 2 ✓ (2)

6.3.1 $y = 4x + 3$ ✓✓ (2)

6.3.2 $y = 4(10) + 3 = 43$ ✓ (1)

6.4 $5x + 4 = 39$ ✓ $\rightarrow 5x = 35$ ✓ $\rightarrow x = 7$ ✓ (3)

6.5 $3 \times 2^2 = 3 \times 4$ ✓ $= 12$ ✓ (2)

[15]

QUESTION 7

7.1 (a) Isosceles, obtuse-angled triangle ✓ (b) Right-angled, isosceles triangle ✓ (2)

7.2 Any TWO: opposite sides are equal ✓; opposite sides are parallel ✓; opposite angles are equal; the diagonals bisect each other (2)

7.3.1 Translation ✓ 6 units to the right ✓ (2)

7.3.2 Reflection ✓ in the line *m* ✓ (2)

7.4 (a) 5 faces ✓ (b) 5 vertices ✓ (c) 8 edges ✓ (3)

7.5 Cube ✓ (1)

[12]

QUESTION 8

8.1 (a) 45 mm ✓ (b) 3,5 km ✓ (c) 0,75 kg ✓ (3)

8.2.1 Large rectangle $10 \times 8 = 80$ cm^2 ✓ minus cut-out $4 \times 3 = 12$ cm^2 ✓ $\rightarrow 68$ cm^2 ✓ (3)
(or split into $6 \times 8 = 48$ and $4 \times 5 = 20 \rightarrow 68$ cm^2)

8.2.2 $6 + 3 + 4 + 5 + 10 + 8$ ✓ $= 36$ cm ✓ (2)

8.3 (a) $6 \times 6 \times 6$ ✓ $= 216$ cm^3 ✓ (b) $3 \times 3 \times 3$ ✓ $= 27$ small cubes ✓ (or $216 \div 8 = 27$) (4)

[12]

QUESTION 9

9.1.1 Soccer ✓ (1)

9.1.2 $^{10}/_{40} \times 100$ ✓ $= 25\%$ ✓ (2)

9.1.3 $^{15}/_{40} \times 360°$ ✓ $= 135°$ ✓ (2)

9.2 (a) HH, HT, TH, TT ✓✓ (one mark for any three correct) (b) $^{1}/_{4}$ ✓ (3)

[8]

TOTAL: 100

END OF MARKING GUIDELINE